UNIT – V

Turbine Performance, Hydraulic Systems & Pumps

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

> Syllabus: Performance of hydraulic turbines (similarity, unit & specific quantities, characteristic curves, governing, selection, cavitation, surge tank, water hammer) ; hydraulic ram, lift, coupling ; fluidics ; centrifugal pumps ; reciprocating pumps.

1. Performance of Hydraulic Turbines

Geometric similarity & model testing

Model and prototype are geometrically similar (same shape, all linear dimensions in a fixed ratio) and operate at dynamically similar conditions (same velocity triangle shapes / same Euler, Reynolds numbers). Turbine models (1:5 – 1:30) are tested to predict the prototype performance.

- Scale ratio $L_r = D_p/D_m$. Similarity laws (same fluid, similar flow) :

$$gH/(N^2D^2) = \text{const}\quad Q/(N D^3) = \text{const}\quad P/(\rho N^3 D^5) = \text{const}$$

- Hence for the same turbine (D fixed) : $N \propto \sqrt{H}$, $Q \propto \sqrt{H}$, $P \propto H^{3/2}$.
- Efficiency scale effect : prototype efficiency is higher than the model's (Moody formula : $(1-\eta_p)/(1-\eta_m) = (D_m/D_p)^{1/5}$).

Unit quantities (for a head of 1 m, same turbine)

$$N_u = N/\sqrt{H}\quad Q_u = Q/\sqrt{H}\quad P_u = P/H^{3/2}$$

Unit speed, unit discharge and unit power are used to compare turbines at one head and to predict performance at another head.

> Ex 1. Model turbine : $H = 6$ m, $N = 400$ rpm, $Q = 0.3$ m³/s, $P = 15$ kW. $N_u = 400/\sqrt{6} = 163.3$; $Q_u = 0.3/2.449 = 0.1225$; $P_u = 15/6^{1.5} = 1.02$. At $H = 36$ m : $N = 163.3 \times 6 =$ 980 rpm ; $Q = 0.1225 \times 6 =$ 0.735 m³/s ; $P = 1.02 \times 36^{1.5} =$ 220 kW.

Specific speed N_s

Speed of a geometrically similar turbine that develops 1 kW under a head of 1 m.

From $P \propto H^{5/2}/N^2$:

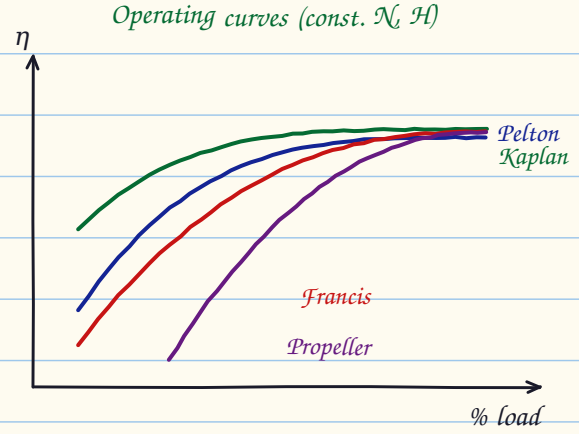
$$N_s = N\sqrt{P} / H^{5/4} \quad (N \text{ rpm}, P \text{ kW}, H \text{ m})$$

- It characterises the **type** of turbine independent of size : Pelton 10 – 60 (single jet) ; Francis 60 – 400 ; Kaplan/propeller 300 – 1000.
- Low N_s ⇒ high head, small discharge. High N_s ⇒ low head, large discharge, high speed.

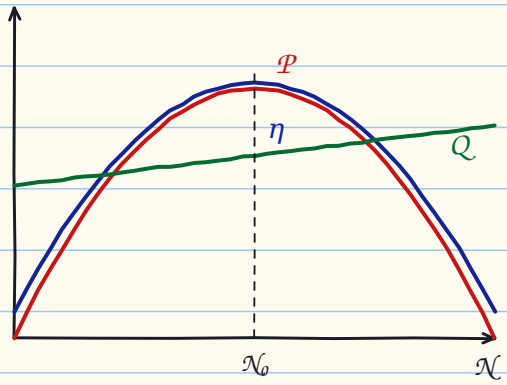
Ex 2. A turbine develops 10 000 kW at 200 rpm under 40 m head : $N_s = 200\sqrt{10000} / 40^{1.25} = 20000/100.6 = 199$ ⇒ Francis turbine.

Characteristic curves

Operating curves (const. N, H)



Main curves (constant head)



- **Main (constant head) curves** : N varied at constant H by changing the load/brake – plots of Q, P and η versus N ; efficiency peaks at the design (unit) speed N_0. For Pelton Q is practically constant (nozzle area fixed).
- **Operating (constant speed) curves** : N and H constant, gate opening varied – η vs % load. Kaplan & Pelton hold η over a very wide range ; Francis falls at part-load ; fixed-blade propeller falls steeply.
- **Constant-efficiency (Muschel / iso-efficiency) curves** : N_u vs Q_u (or H vs Q) contours of equal η ; the hill-diagram gives the full operating range and best-efficiency zone.

Governing of turbines

Maintaining a constant speed under varying load for the alternator frequency (50 Hz). A governor senses speed change and adjusts discharge through servo-motors (oil

pressure).

Turbine	Governing method	Remark
Pelton	Spear (needle) valve ; jet deflector / auxiliary nozzle	deflector avoids water hammer when load drops suddenly
Francis	Guide vane (wicket gate) opening	by-pass valve in extreme cases
Kaplan	Guide vanes + runner blade pitch	double regulation

Selection of the type of turbine

Head (m)	Type	Reason
> 250	Pelton (impulse)	high head, low flow, N_s low
60 – 250	Francis	medium head & flow
< 60	Kaplan / propeller	large discharge, high N_s
Fluctuating load	Pelton, Kaplan	flat efficiency curve
Very small heads (< 5 m)	Bulb / tubular turbines, water wheels	axial flow, run-of-river

Other factors : available discharge, cost, floor space, tail-water variation, cavitation limit, speed of generator, maintenance.

2. Cavitation in Turbines

When the local absolute pressure falls to the vapour pressure of water, vapour bubbles form (boiling at ordinary temperature) and are carried to a high-pressure region where they collapse suddenly, producing shock pressures (up to 1000 atm).

- **Where?** Reaction turbines : on the back (suction side) of runner blades near the outlet, draft tube inlet ; Pelton : at the nozzle tip and bucket splitter (rarely).
- **Effects** : pitting and erosion of the metal, noise & vibration, loss of power, drop in efficiency, reduced life.
- **Thoma's cavitation parameter** :

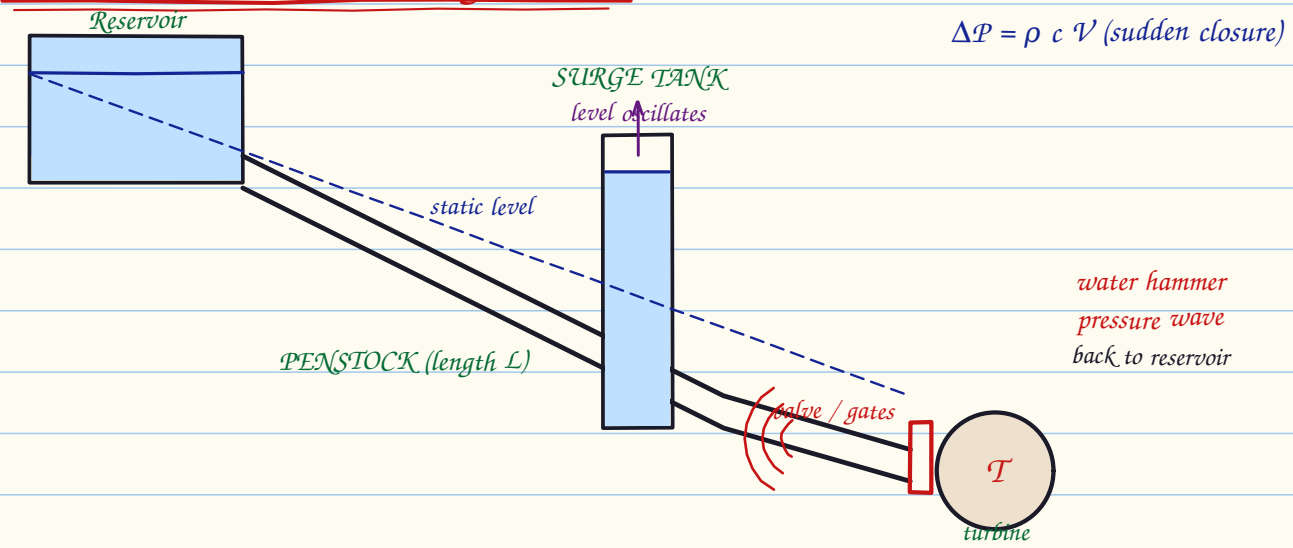
$$\sigma = (H_{atm} - H_v - H_s) / H = NPSH / H$$

- H_{atm} = atmospheric head (10.3 m water), H_v = vapour pressure head, H_s = suction (setting) height of runner outlet above tail water. The turbine is free from cavitation if $\sigma > \sigma_c$ (critical value from model tests, depends on N_s : $\sigma_c \approx 0.0432$

$(N_s/380.78)^2$ for Francis).

- Hence the maximum safe setting : $H_s \le H_{atm} - H_v - \sigma_c H$. For high-head Francis the runner must be set below tail water (negative H_s).
- **Prevention** : keep runner setting low, use cavitation-resistant materials (stainless steel, stellite), polished surfaces, correct draft-tube design, air admission, avoid part-load operation, good blade shapes.

3. Water Hammer & Surge Tank



Water hammer

When the flow in a long pipe is suddenly retarded (valve closed / gates shut on load rejection), the kinetic energy changes into pressure energy – pressure waves travel to and fro (speed $c \approx 1000$–1400 m/s) producing a severe pressure rise that may burst the pipe.

- Wave speed : $c = \sqrt{(K/\rho)} / \sqrt{(1 + K D/(E t))}$ (K bulk modulus, E Young's modulus of pipe, t thickness).
- **Sudden closure** (closing time $t_c \le 2L/c$) : rise in pressure $\Delta P = \rho c V$ (Joukowsky). **Gradual closure** ($t_c > 2L/c$) : $\Delta P \approx \rho L V / t_c$.
- Phases : (i) compression wave travels to reservoir (time L/c), (ii) reflected as release wave, (iii) negative pressure at the valve, (iv) returns ; period $4L/c$.

> Ex 3. Penstock $L = 1000$ m, $V = 2$ m/s, $c = 1200$ m/s : $\Delta P = 1000 \times 1200 \times 2 = 2.4$ MPa (≈ 245 m of water). Critical time $2L/c = 1.67$ s.

- **Prevention** : slow closing valves, surge tank, relief/pressure-regulating valves, air vessels, thick pipes, deflector in Pelton.

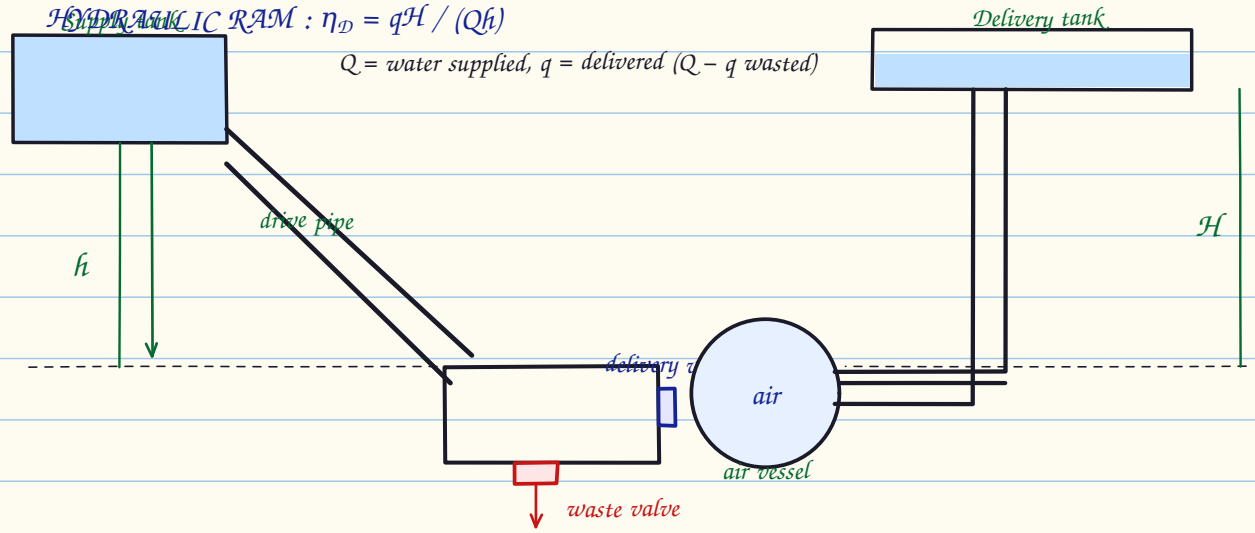
Surge tank

An open-top cylindrical storage tank connected to the penstock near the turbine.

- Functions : (a) during load rejection, absorbs the extra water (level rises) – prevents water hammer reaching the tunnel/reservoir ; (b) during load increase, supplies the extra water until the supply in the conduit is accelerated – so turbine isn't starved ; (c) reduces water-hammer pressure in the penstock section.
- Types : simple, restricted-orifice, differential (Johnson), closed (air-cushion), and with spillway. Water level in the surge tank oscillates (mass oscillation) and is damped by friction.

4. Hydraulic Systems

(a) Hydraulic ram



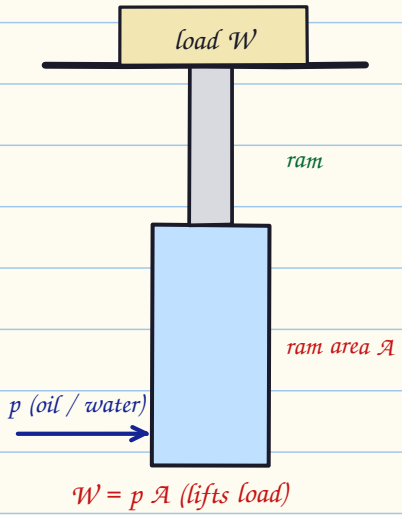
A device that lifts a small quantity of water to a great height using the energy of a large quantity falling through a small height – it works on the water hammer principle and needs no external power.

- Working : water flows from the supply tank down the drive pipe and out through the open waste (impulse) valve, accelerating. When velocity is large the waste valve snaps shut → sudden stoppage → pressure rise → opens the delivery valve and forces water into the air vessel and up the delivery pipe. Pressure falls, delivery valve closes, waste valve reopens and the cycle repeats (60 – 100 beats/min).
- Air vessel gives a steady delivery ; a snifter valve supplies the air it dissolves.

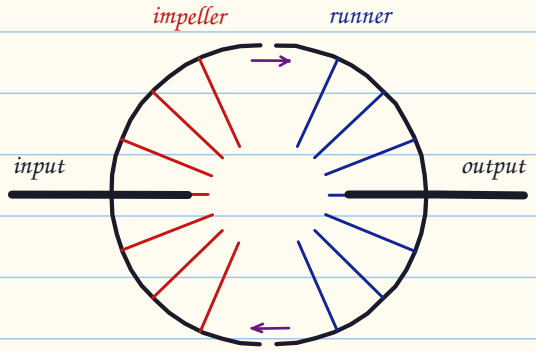
- Efficiency : D'Aubuisson $\eta_D = q\,H / (Q\,h)$; Rankine $\eta_R = q\,(H - h) / ((Q - q)\,h)$ where h = supply head, H = delivery head, Q = total flow from supply, q = delivered flow.

> Ex 4. $h = 2$ m, $H = 10$ m, $Q = 150$ l/min, $q = 10$ l/min : $\eta_D = (10 \times 10)/(150 \times 2) = 33$ % ; $\eta_R = 10(10 - 2)/(140 \times 2) = 28.6$ %.

(b) Hydraulic lift



FLUID COUPLING ($T_{in} = T_{out}$, slip 2–5 %)



- Used in buildings, garages, service stations. Liquid (oil) under pressure p acts on the ram of area A and lifts the cage : $W = p\,A$ (with pump pressure of 10–15 MPa a small ram lifts heavy loads — Pascal's law).
- Types : ***direct-acting*** (ram directly under the car ; travel = ram stroke), ***suspended*** (ram with pulley and chains – car travels farther than the ram, speed ratio 2:1 etc.), counter-weighted lifts for tall buildings. Pump, oil tank, check valves and control valves complete the system.

(c) Hydraulic coupling (fluid coupling)

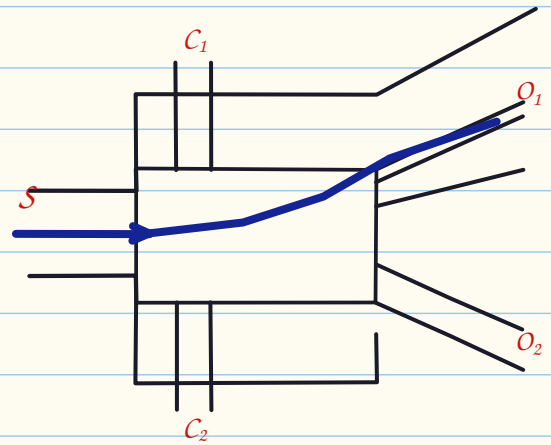
- Two identical halves – ***impeller*** (driving, pump) connected to the engine and ***runner*** (driven, turbine) connected to the load – each like a half toroid with radial vanes, filled with oil, with no mechanical contact. Oil thrown outward by the impeller enters the runner at the rim, transfers its angular momentum and returns inward.
- Torque input = torque output (no torque multiplication) ; slip $s = (N_1 - N_2)/N_1 \approx 2 - 5$ % ; efficiency $\eta = N_2/N_1 \approx 95 - 98$ %.
- Advantages : smooth start, cushions shocks, protects against overload, allows motor to reach full speed before load picks up. Used in buses, locomotives,

conveyors, ship propulsion.

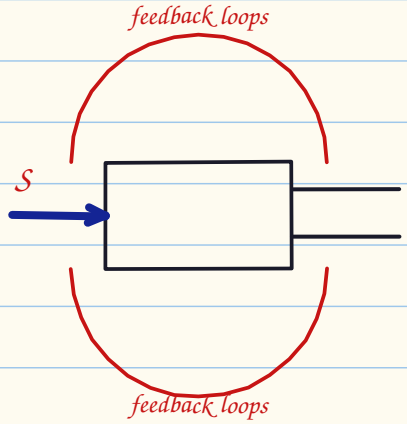
- **Torque converter** adds a fixed stator (reaction vanes) so that output torque exceeds input torque (multiplication up to 2–3×) – automatic transmissions. Other systems : hydraulic accumulator, intensifier, jacks, crane, press, brake.

5. Fluidics

(a) Bistable wall-attachment amplifier



(b) Fluidic oscillator



Fluidics = technology of using fluid flow (liquid/gas jets) for sensing, logic, amplification and control – devices with no moving parts.

- **Fluidic amplifiers** : a power jet from a supply nozzle S passes through an interaction chamber to two output channels. A small control flow at C_1 or C_2 deflects the jet to the other output. Types : bistable (wall-attachment, Coanda effect) – jet latches to a wall – flip-flop memory ; proportional – output varies with control flow ; turbulence amplifier – laminar jet triggered to turbulence ; vortex amplifier.
- **Fluidic sensors** : back-pressure sensor (proximity), interruptible-jet sensor (object breaks the jet), fluidic flow meter (vortex shedding), temperature (speed of sound / oscillation frequency), pressure sensors, level sensors.
- **Fluidic oscillators** : output jet fed back through feedback loops to the control ports, producing a continuous switching at frequency depending on loop length ($f \propto 1/\text{delay}$) – used as flow-meters, sprinklers, windshield washers, flow-modulation.

Advantages	Limitations
No moving parts – high reliability, long life, low maintenance	Low speed of response (few ms – kHz range limited by sound speed)

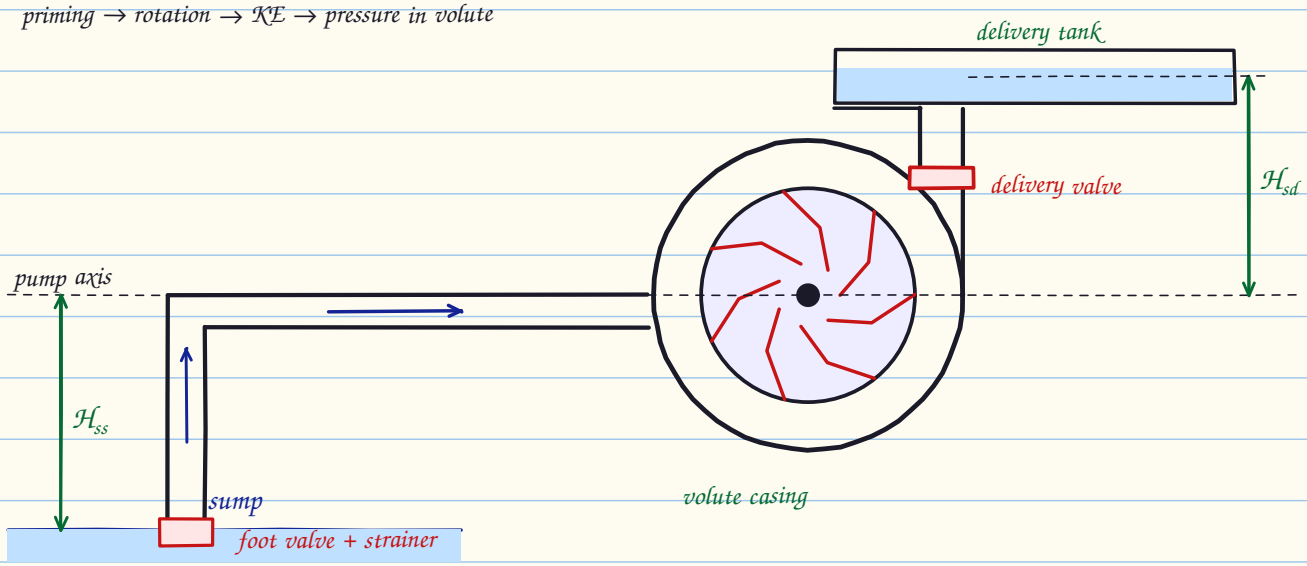
Advantages	Limitations
Work in hostile environments : high temp, radiation, explosive, vibration	Low gain & low signal levels ; need fluid supply and power continuously
Cheap, simple construction (moulded / etched plates)	Sensitive to dirt, viscosity & supply pressure changes ; noise
Fail-safe, inherently explosion-proof	Slower than electronics ; interfacing with other systems needed

Applications : logic and sequence control in factories, aircraft/missile control surfaces, fuel and engine control, flow measurement, medical (respirators, ventilators), automatic washers, vending, process industries, nuclear plants.

6. Centrifugal Pumps

CENTRIFUGAL PUMP

priming → rotation → KE → pressure in volute



A rotodynamic machine converting mechanical energy into hydraulic energy – the reverse of a radial-flow reaction turbine. Impeller rotating in a casing gives the water kinetic energy (centrifugal head) which is converted to pressure in the volute/diffuser.

Working & parts

- **Priming** : filling the suction pipe and casing with the liquid before starting – a centrifugal pump cannot lift air. As the impeller rotates, the liquid is thrown outward by centrifugal force, creating low pressure at the eye ; atmospheric pressure pushes more liquid up the suction pipe.
- Parts : impeller (backward-curved vanes), casing (volute, vortex, or diffuser type), suction pipe with foot valve and strainer, delivery pipe with delivery (regulating)

valve, shaft and bearings, stuffing box.

Classification

- By number of impellers : single-stage / multi-stage (for high head) ; by suction : single / double suction ; by impeller : open, semi-open, closed ; by casing : volute, vortex (with whirlpool chamber), diffuser (turbine pump).
- By head : low (< 15 m), medium (15 – 40 m), high (> 40 m) ; by specific speed : radial, mixed, axial ; by shaft : horizontal / vertical ; by priming : self-priming / ordinary.

Heads

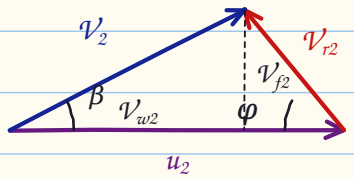
- Suction head H_s (centre-line above sump) ; delivery head H_d ; **static head** $H_{st} = H_s + H_d$.

$$H_m = H_{st} + h_{fs} + h_{fd} + V_d^2/2g \text{ (manometric head)}$$

- Manometric head = head imparted by impeller – losses inside pump = $(p_d - p_s)/\rho g + (V_d^2 - V_s^2)/2g + (z_d - z_s)$.

Work done by the impeller

OUTLET velocity triangle



Euler head per unit weight :

$H_e = (V_{w2}u_2 - V_{w1}u_1)/g$

radial entry ($\alpha = 90°$) : $V_{w1} = 0$

$V_{w2} = u_2 - V_{f2}\cot\varphi$

$\varphi < 90°$ backward curved vanes (usual)

$Q = \pi D_2 B_2 V_{f2}$

$$\text{Work/weight} = (V_{w2}u_2 - V_{w1}u_1)/g \text{ radial entry : } V_{w2}u_2/g$$

- Power given to water = $\rho Q(V_{w2}u_2 - V_{w1}u_1)$; $Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$.

Efficiencies

Efficiency	Definition	Typical
Manometric η_{man}	$H_m / (V_{w2}u_2/g) = gH_m/(V_{w2}u_2)$	65 – 90 %
Mechanical η_m	power at impeller / shaft power	90 – 95 %
Volumetric η_v	$Q/(Q + \text{leakage})$	95 – 98 %
Overall η_o	$\rho g Q H_m / SP = \eta_{man}\,\eta_m\,\eta_v$	60 – 85 %

- **Minimum starting speed** : delivery begins only when the centrifugal head ≥ manometric head : $(u_2^2 - u_1^2)/2g \geq \mathcal{H}_m \Rightarrow \mathcal{N}_{min} = 120\ \eta_{man}\ \mathcal{V}_{w2}\ \mathcal{D}_2 / [\pi\ (\mathcal{D}_2^2 - \mathcal{D}_1^2)]$.

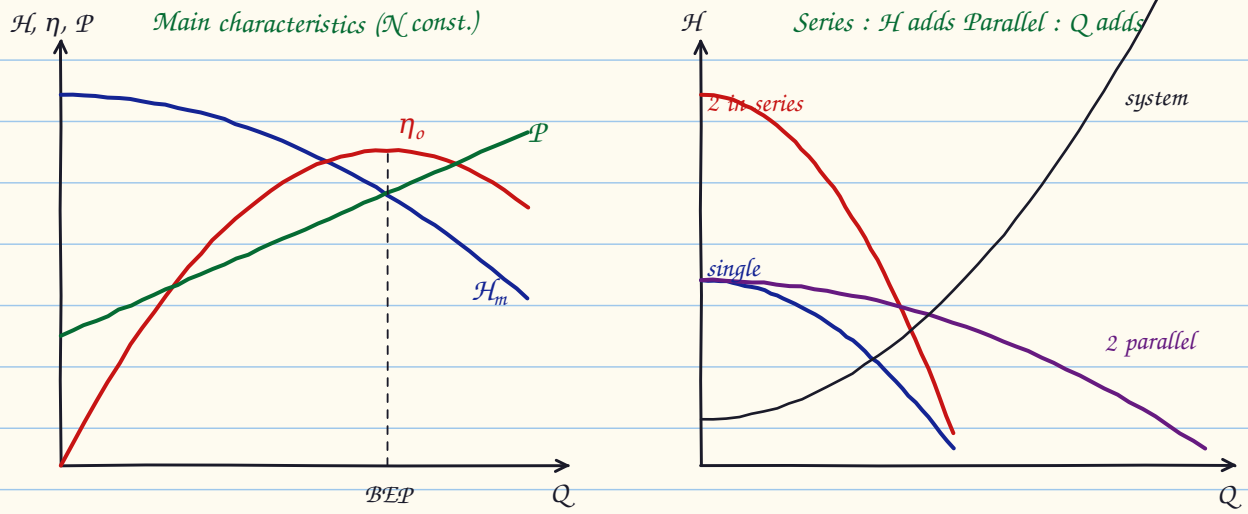
Ex 5. $\mathcal{D}_2 = 0.4$ m, $\mathcal{N} = 1450$ rpm, $\mathcal{V}_{f2} = 2.5$ m/s, $\varphi = 30°$, $\eta_{man} = 0.8$. $u_2 = \pi(0.4)(1450)/60 = 30.4$ m/s ; $\mathcal{V}_{w2} = u_2 - \mathcal{V}_{f2}/\tan\varphi = 30.4 - 4.33 = 26.0$ m/s ; $\mathcal{H}_m = \eta_{man}\ \mathcal{V}_{w2} u_2/g = 0.8 \times 26.0 \times 30.4/9.81 =$ 64.5 m.

Specific speed

$$\mathcal{N}_s = \mathcal{N}\sqrt{Q}\ /\ \mathcal{H}_m^{3/4} \quad (\mathcal{N}\ \text{rpm},\ Q\ m^3/s,\ \mathcal{H}_m\ m)$$

Speed of a geometrically similar pump that delivers 1 m^3/s against 1 m head. Radial (centrifugal) ≈ 10 – 80 ; mixed flow 80 – 160 ; axial (propeller) > 160. Similarity laws $\mathcal{H}/(\mathcal{N}^2\mathcal{D}^2)$, $Q/(\mathcal{N}\mathcal{D}^3)$, $\mathcal{P}/(\mathcal{N}^3\mathcal{D}^5)$ constant.

Pumps in series & parallel



- **Series** : delivery of first pump feeds the suction of second ⇒ same Q, heads add ($\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2$) – used for very high heads (or multi-stage).
- **Parallel** : each pump draws from the sump and delivers to the same main ⇒ same $\mathcal{H}$, discharges add ($Q = Q_1 + Q_2$) – used for large discharge. With a steep system curve, two parallel pumps give less than twice the flow.

Performance characteristic curves

- **Main curves** (constant $\mathcal{N}$) : $\mathcal{H}_m$, $\mathcal{P}$ and η vs Q. $\mathcal{H}$ falls with Q, $\mathcal{P}$ rises, η rises to a maximum – the best efficiency point (BEP) – then drops. At $Q = 0$ (shut-off) power is least : start with delivery valve closed.

- **Operating curves** (constant head & efficiency ($\mathcal{N}$ varied)) ; **constant-efficiency curves** (Muschel) from tests at various speeds. **System curve** $\mathcal{H} = \mathcal{H}_{st} + K Q^2$; the intersection with the pump curve is the operating point.

Cavitation & NPSH

- Low suction pressure at the impeller eye (high lift, high speed, hot liquid) ⇒ cavitation ⇒ noise, vibration, fall of $\mathcal{H}$ & η, pitting.

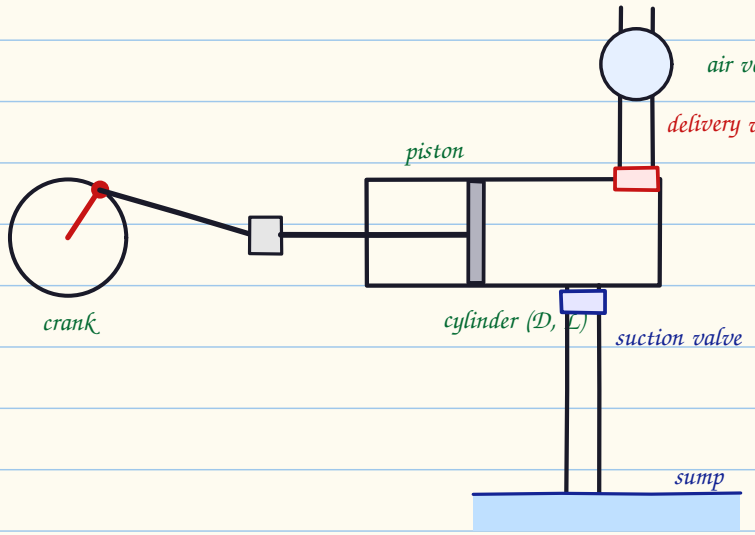
$$\mathcal{NPSH}_a = \mathcal{H}_{atm} - \mathcal{H}_v - \mathcal{H}_s - h_{fs} \text{ require } \mathcal{NPSH}_a > \mathcal{NPSH}_r$$

- $\mathcal{NPSH}_r$ is given by the manufacturer (test) ; Thoma's factor $\sigma = \mathcal{NPSH}_r/\mathcal{H}_m$. Remedies : reduce suction lift / friction, larger suction pipe, lower speed, double-suction impeller, inducers, cool the liquid.

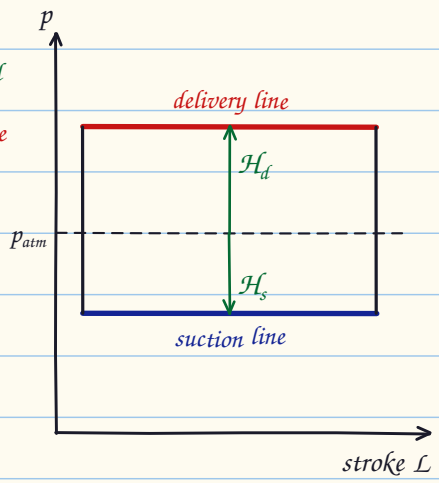
Ex 6. Sea-level pump : $\mathcal{H}_{atm}$ = 10.3 m, water at 30°C $\mathcal{H}_v$ = 0.43 m, suction lift 3 m, h_{fs} = 0.5 m : $\mathcal{NPSH}_a$ = 10.3 − 0.43 − 3 − 0.5 = 6.37 m.

7. Reciprocating Pumps

RECIPROCATING PUMP (single acting)



Indicator diagram (ideal)



Positive displacement pumps – the piston moving in a cylinder alternately draws liquid through the suction valve and forces it out through the delivery valve. Power to piston through a crank and connecting rod from a motor/engine.

Working & types

- Suction stroke : piston moves out, pressure in cylinder falls below atmospheric, suction valve opens, liquid enters ; delivery valve closed. Delivery stroke : piston moves in, suction valve closes, delivery valve opens.

- Types : **single-acting** (one delivery per revolution), **double-acting** (two), **duplex / triplex** (2 or 3 cylinders for smoother flow) ; **air vessels** on suction and delivery pipes.

Discharge, slip & power

$$Q_{th} = A\,L\,N / 60 \text{ (single)} \quad Q_{th} = 2\,A\,L\,N / 60 \text{ (double, neglecting rod)}$$

- Actual discharge $Q_{act} < Q_{th}$ owing to leakage, valve delay. **Slip** $= Q_{th} - Q_{act}$; % slip $= (Q_{th} - Q_{act})/Q_{th} \times 100$; coefficient of discharge $C_d = Q_{act}/Q_{th}$. *Negative slip* occurs if the delivery valve opens late / suction valve closes late (long suction pipe, high speed) – $Q_{act} > Q_{th}$.

$$\text{Work done/s} = \rho\,g\,Q\,(H_s + H_d) \quad \text{Power} = \rho\,g\,A\,L\,N\,(H_s + H_d) / 60 \text{ (single-acting)}$$

Ex 7. Single-acting : D = 150 mm, L = 300 mm, N = 60 rpm, H_s = 4 m, H_d = 20 m, Q_{act} = 5.0 l/s. $Q_{th} = 0.017671 \times 0.3 \times 60/60 = 5.30$ l/s ; slip = (5.30 – 5.00)/5.30 = 5.7 % ; power = 9810 × 0.0053 × 24 = 1.25 kW.

Indicator diagram

Plot of pressure in the cylinder (ordinate) against piston displacement (abscissa) – drawn by an indicator on the cylinder.

- **Ideal diagram** : a rectangle – suction line at $(H_{atm} - H_s)$ and delivery line at $(H_{atm} + H_d)$. Area = work done per cycle.
- **Effect of acceleration** : at the beginning of each stroke the piston accelerates, the liquid column must also be accelerated ⇒ extra pressure. *Acceleration head* $h_a = (L_p/g)(A/a)\,\omega^2\,r\cos\theta$ (maximum at beginning $= (L_p/g)(A/a)\omega^2 r$) ; at the end of stroke it is negative – pressure at the end of delivery stroke may fall to the vapour pressure ⇒ *separation* (cavitation) in the suction pipe.
- **Effect of friction** : friction head $h_f = (4\,f\,L_p/(2gD))\,(A\,v/a)^2$ varies as the square of the velocity, maximum in the middle of stroke ⇒ indicator diagram bulges. Actual diagram therefore has sloping lines and an area larger than the ideal.
- **Air vessels** : placed close to the cylinder on both suction and delivery sides ; they (i) smooth the flow, (ii) reduce acceleration head (by shortening the length of liquid that must be accelerated) and friction head, (iii) permit higher speed without separation, (iv) save power (≈ 39 % single-acting delivery side).

Reciprocating vs centrifugal pump

Item	Reciprocating	Centrifugal
Discharge	small, pulsating, fixed by speed	large, continuous, varies with head
Head	very high	low–medium (multi-stage for high)
Efficiency	high at all heads	max near BEP only
Priming	self-priming	needs priming
Size, cost, wear	large, heavy, many parts	compact, light, simple, cheap
Liquids	clean liquids ; viscous OK	clean, low-viscosity ; handles solids if open impeller

8. Quick Formula Sheet – Unit V

Item	Formula
Unit quantities	$N_u = N/\sqrt{H}$; $Q_u = Q/\sqrt{H}$; $P_u = P/H^{3/2}$
Specific speed (turbine)	$N_s = N\sqrt{P}/H^{5/4}$
Thoma factor	$\sigma = (H_{atm} - H_v - H_s)/H$
Water hammer	$\Delta P = \rho c V$ (sudden) ; $\rho L V/t$ (gradual)
Hydraulic ram	$\eta_D = qH/(Qh)$
Pump manometric head	$H_m = H_{st} + h_f + V_d^2/2g$
Pump efficiency	$\eta_o = \rho g Q H_m/SP$
Specific speed (pump)	$N_s = N\sqrt{Q}/H_m^{3/4}$
NPSH	$NPSH_a = H_{atm} - H_v - H_s - h_{fs}$
Reciprocating	$Q = ALN/60$; slip $= (Q_{th} - Q_{act})/Q_{th}$

9. Important Questions

- Define unit speed, unit discharge, unit power and specific speed of a turbine; derive N_s. Explain characteristic curves.
- Explain governing of Pelton, Francis & Kaplan turbines. How is a turbine selected for a given site?
- What is cavitation? Effects, Thoma's parameter, prevention in turbines and pumps.

- Explain water hammer & surge tank with sketches. Derive the pressure rise for sudden closure.
- Describe the hydraulic ram, hydraulic lift and fluid coupling with neat sketches; define efficiency of a ram.
- What is fluidics? Explain amplifier, sensor and oscillator; advantages, limitations & applications.
- Explain the working of a centrifugal pump; derive work done, manometric head, efficiencies, minimum starting speed, specific speed.
- Pumps in series/parallel; characteristic curves; NPSH. Reciprocating pump – discharge, slip, indicator diagram, air vessels.